



FEEDBACK

YOUR SUGGESTIONS

EFY DVD

In EFY + DVD section on page 66 of March issue, the Arduino IDE 1.7.5 mentioned seems to be wrong. The latest Arduino version is 1.6.8 only.

Raju P. Abraham

Palakkad, Kerala

EFY. Version 1.7.5 can be seen on <http://www.arduino.org/> and the difference between it and version 1.6.8 can be checked on <http://arduino.stackexchange.com/questions/11824/difference-between-arduino-cc-and-arduino-org>

MOTION DETECTOR

Regarding 'Motion Detector Using NE555 Timer' circuit published in August 2015 issue, can you help me with the equation for the monostable used in this circuit: How 'R' and 'C' are related with the time?

Anjali

Through email

The author Kumar Abhishek replies:
The equation for the time period of monostable circuit is given by:

$$\text{Time } (\tau) = 1.1(R \times C)$$

where τ is time in seconds, R is resistance in ohms and C is capacitance in Farads.

Here, R is connected to the 7th pin of NE555. The actual time may be somewhat different from the calculated value due to the tolerances in the values of the components.

POWER SAVER

I am interested in 'Infrared Sensor Based Power Saver' DIY article published in March issue. What type of connectors are used in the PCB?

Shruti Choudhary

Through email

The author Fayaz Hasan replies:

I used strip connectors for CON1 and CON3, burg strip with 3-wire F/F for CON2 (PIR) and 2-wire reimate connector for transformer X1. But, to use on PCB, you can use 3-pin reimate connectors for CON1 and CON3, burg strip for CON2 (PIR) and either burg strip or reimate of 2-pin connector for X1. All have 2.54mm (0.1-inch) pitch to suit the PCB. To simplify the 230V AC connections, you can control only the phase or line L through relay RL1 and directly

connect neutral N (and earth, if any) to the load.

USB CONTROLLED POWER

In 'Automatic USB Controlled Power Switch' DIY article published in November 2015 issue, I did not get the exact output. Is LED TV essential for this project, or any other device will do?

Sachin Kurian

Through email

The author Ajith Kumar replies:

You can use this circuit to control any device having a USB output socket such as laptop, set-top-box, home theatre, etc.

CAR BATTERIES

In 'Device for Maintaining Car Batteries' DIY article published in August 2015 issue, I have doubts in the use of IC 7805, a 5V voltage regulator, when the output of the circuit required is about 15.5V.

Sang Nguyen

Through email

The author Petre Tzv Petrov replies:

In this circuit, 7805 works as a current generator and not in its classical application of series regulator with 5V output between the output of the regulator and the ground of the circuit.

Please see the data sheet of 7805. The output current of IC1 is set with R2, R3 and R4. The output voltage over the battery 1 (BATT.1) is limited by the series network of LED1, ZD1 and ZD2. You can use any appropriate combination of LEDs, diodes, Schottky diodes and Zener diodes to limit the output voltage to the required value. The circuit works well after proper montage is done.



FROM FACEBOOK

EFY's Electronics Design Community

Nice to see 'What Drives Power Conversion Components' article in March edition. Very relevant article for all the electronics engineers in my list!

Ganesh Shankar

EFY. Thanks for the feedback!

'Spot An Error' AWARD Winners

In '4-Channel Multi-Mode Audio Amplifier' DIY article published in March issue, in Fig. 4, capacitors C9-C12 and C15-C18 should be 2200µF, 25V and 2200µF, 35V respectively as shown in Fig. 1 and parts list.
Tanmay Dasgupta

In 'Vibration-Activated Smart CRO Probe' DIY article published in March issue, the live (L) and neutral (N) points are wrongly mentioned in the component layout. Switch S1 is wrongly mentioned in the parts list as SPDT switch instead of SPST switch.
Samiuddin